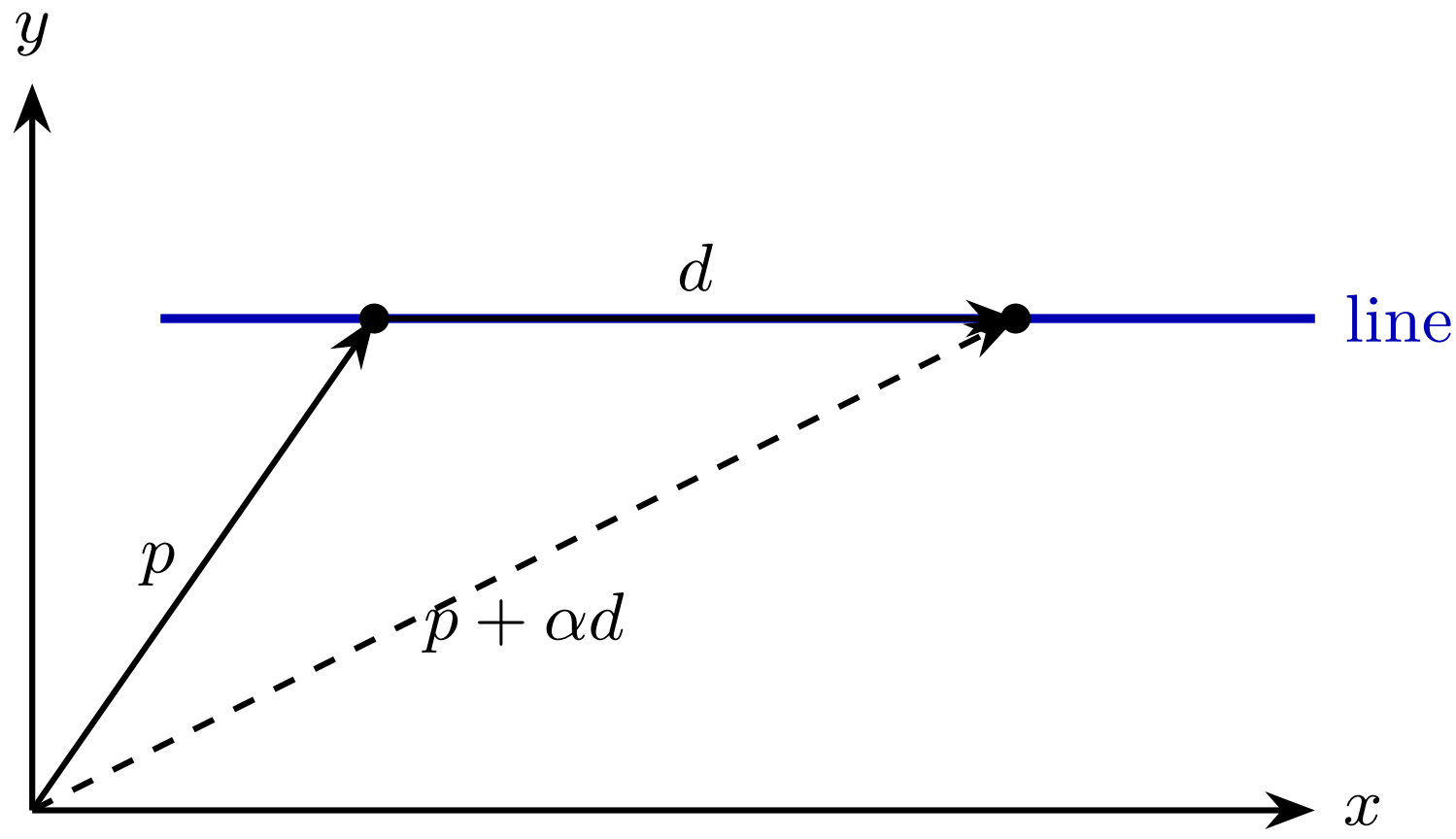

Changing the chosen point along the line does not change $m = p \times d$, because $d \times d = 0$.